

PERSONAL INFORMATION

Georgios Kissas
OAT X11, Andreasstrasse 5, 8092 Zürich, Switzerland
☎ +41 789699577
✉ gkissas@ai.ethz.ch
in www.linkedin.com/in/georgioskissas/
📄 scholar.google.com/georgioskissas



AWARDS

- **Gold Reviewer ICML**, 2026
- **EUROMECH Colloquium 662** Early Career Invited Talk Award, 2026
- **AWS Free Credit Trainium TPU Servers**, 2025
- **Top Reviewer NeurIPS**, 2024
- **Asuera Stiftung Scholarship** ETH Zürich Foundation, 120000 CHF, 2023
- **Post Doctoral Fellowship** ETH AI Center (1.5% acceptance rate), 2023
- **Outstanding TA Award** MEAM UPenn, 2021
- **Gerondellis Foundation** Scholarship Award, 2020
- **MIT Travel Grant** Learning for Dynamical Systems and Control, 2019
- **Doctoral Fellowship Award** University of Pennsylvania, 2018
- **Limmat Stiftung Award** Award for Academic Excellence, 2017

PROFESSIONAL EXPERIENCE

- Senior Data Scientist, Research** Swiss Data Science Center *November 2025–Present*
- Development of novel deep learning algorithms for scientific discovery using language modeling, large scale AI with applications to Exascale Cosmological Astrophysics simulations, and industrial CFD applications.
- Post Doctoral Researcher & Lecturer** ETH Zurich *June 2023–November 2025*
- Joint Post Doctoral positions between the Applied Mathematics, Engineering and Computer Science.
 - Development of novel deep learning algorithms for scientific discovery through language modeling, and first-principles AI with applications to dynamical systems.
- Research Assistant** University of Pennsylvania, PI Lab *August 2018–May 2023*
- Worked in the areas of Deep Learning for physics simulations and precision medicine, Operator Learning, Fluid Dynamics for Cardiovascular Flows, Bayesian Inference, Deep Learning for Dynamical Systems and Sensitivity Analysis.
- Technische Angestellte** Wacker Chemie AG *June 2018–August 2018*
- Developed and enhanced analytical and computational models for modeling solid liquid two-phase flows and computing the dimensions of fractal nanoparticles and the effective viscosity of suspensions.
- Research Assistant** National Technical University of Athens, COMSE Group *May 2016–May 2018*
- Developed a Finite and Spectral Element solvers for the system of equations derived from Self-Consistent Field Theory.

EDUCATION

- University of Pennsylvania**, Philadelphia, PA, USA
- PhD, Mechanical Engineering and Applied Mechanics, May 2023
Thesis: *"Towards digital twins for cardiovascular flows: A Hybrid Machine Learning and Computational Fluid Dynamics Approach"*
Advisor: Paris Perdikaris
- National Technical University of Athens**, Athens, Greece
- M.S., Computational Mechanics (Fluids), May, 2017, (2nd Highest GPA)
Thesis: *"A computational study of Self-Consistent Field Theory for polymer interfaces"*
Advisor: Doros Theodorou
- University of Patras**, Patras, Greece
- Diploma, Mechanical and Aeronautical Engineering, February, 2015 (Honors)
Thesis: *"A comparative study of Wavelet and Hilbert-Huang Transformations for fault detection in slew ring bearings"*

PATENTS

Computer Systems and Methods for Learning Operators, P. Perdikaris, G.J. Pappas, J. Seidman, **G. Kissas** - Provisional Patent Application 22-9935 (<https://www.freepatentsonline.com/y2023/0214661.html>).

PUBLICATIONS

- Brivio, S, **Kissas, G.**, Manzoni, A., and Mishra, S., 2026, Reduced-Order Operator Learning through Perceivers: Dissecting the Hidden Mechanisms. (*To appear in the ArXiv*)
- Koren, N., Hofmann, T., **Kissas, G.**, METRO: Metric-Enhanced Token Routing Operator. (Under Review NeurIPS 2026)
- Koren, N., Hofmann, T., **Kissas, G.**, Advectra: Asymmetric Latent Transport for Non-Stationary Physics. (Under Review NeurIPS 2026)
- Kissas, G.**, Bonev, B., Oikonomou, O., Lingsch, L., Chamon, F. O. L., 2026, Attention, Generators, and Semigroups: Structurally Stable Deep Transformers. (Under Review NeurIPS 2026)
- Yu, K., Chatzi, E. and **Kissas, G.**, 2026. Neuro-Symbolic ODE Discovery with Latent Grammar Flow. arXiv preprint, arXiv:2604.16232.
- Lingsch, L., **Kissas, G.**, Jakubik, J., Mishra, S., 2026, Phaedra: Learning High-Fidelity Discrete Tokenization for the Physical Science, arXiv preprint, arXiv:2602.03915
- Oikonomou, O., Lingsch, L., Grund, D., Mishra, S. and **Kissas, G.**, 2025. Neuro-Symbolic AI for Analytical Solutions of Differential Equations. Forty-Third International Conference on Machine Learning, 2026
- Renner, D. Mishra, S., and **Kissas, G.**, 2026. Accelerated Patient-Specific Calibration via Differentiable Hemodynamics Simulations. arXiv preprint arXiv:2412.14572.(*Under Review ACM TOMS*)
- Yu, K., Chatzi, E. and **Kissas, G.**, 2025. Grammar-based Ordinary Differential Equation Discovery. Mechanical Systems and Signal Processing, 240, p.113395.
- Sokolow, R., Bhattacharya, A., **Kissas, G.**, Thompson, E.W., Beeche, C., Swago, S., Zhang, D., Viswanadha, M., Morse, C., Chirinos, J. and Damrauer, S., 2025. Reduced order computational fluid dynamic simulations in the thoracic aorta are associated with disease recorded in a medical biobank. Scientific Reports, 15(1), p.39203.
- Lingsch, L.E., Grund, D., Mishra, S. and **Kissas, G.**, 2024. FUSE: Fast Unified Simulation and Estimation for PDEs, *Thirty-eighth Conference on Neural Information Processing Systems*, 2024.
- Kissas, G.**, Mishra, S., Chatzi, E. and De Lorenzis, L., 2024. The Language of Hyperelastic Materials. *Computer Methods in Applied Mechanics and Engineering*, 428, p.117053.
- Seidman, J.H., **Kissas, G.**, Pappas, G.J. and Perdikaris, P., 2023. Variational Autoencoding Neural Operators, *International Conference on Machine Learning*, 2023.
- Kissas, G.***, Seidman *, J.H., Guilhoto, L.F., Preciado, V.M., Pappas, G.J. and Perdikaris, P., 2022. Learning Operators with Coupled Attention. *Thirty-sixth Conference on Neural Information Processing Systems, Journal to Conference Track (Spotlight)*
- Seidman, J.H.*, **Kissas, G. ***, Perdikaris, P. and Pappas, G.J., 2022. NOMAD: Nonlinear Manifold Decoders for Operator Learning, *Thirty-sixth Conference on Neural Information Processing Systems. (Featured in the "Paper of the Week" section of DeepAI)*
- Kissas, G.***, Hwuang, E., Thompson, E.W., Schwartz, N., Detre, J.A., Witschey, W.R. and Perdikaris, P., 2022. Feasibility of Vascular Parameter Estimation for Assessing Hypertensive Pregnancy Disorders. *Journal of Biomechanical Engineering*.

Yang, Y., **Kissas, G.**, Perdikaris, P., 2022. Scalable uncertainty quantification for deep operator networks using randomized priors. *Computer Methods in Applied Mechanics and Engineering*, 399, p.115399.

Kissas, G.*, Seidman *, J.H., Guilhoto, L.F., Preciado, V.M., Pappas, G.J. and Perdikaris, P., 2022. Learning Operators with Coupled Attention. *Journal of Machine Learning Research*, 23(215), pp.1-63

Kissas, G., Yang, Y., Hwuang, E., Witschey, W.R., Detre, J.A. and Perdikaris, P., 2020. Machine learning in cardiovascular flows modeling: Predicting arterial blood pressure from non-invasive 4D flow MRI data using physics-informed neural networks. *Computer Methods in Applied Mechanics and Engineering*, 358, p.112623. (**Featured in the "Most Cited Articles" section of CMAME as one of the most cited papers of the years 2019-2023.**)

* indicates equal contribution

WORKSHOPS AI for Science Track for AI+X Summit: Georgios Kissas with Vera Balmer and Sid Mishra, 2024. (<https://www.plusx.ai>)

2022 TRIPODS Winter School and Workshop on Interplay between Artificial Intelligence and Dynamical Systems: Paris Perdikaris with Georgios Kissas and Jacob Seidman, Supervised learning in function spaces, Mathematical Institute for Data Science, Johns Hopkins University, 2022. (https://github.com/PredictiveIntelligenceLab/TRIPODS_Winter_School_2022)

MINISYMPOSIA "Biophysics-Informed Machine Learning", Organizer/Chair: Georgios Kissas, Co-chair: Jacob Seidman, 2023 Platform for Advanced Scientific Computing (PASC) Conference, Davos, Switzerland, 2023.

ACADEMIC ETH, Zurich, Switzerland
EXPERIENCE Lecturer

- Projects in Machine Learning Research, Spring 2025.
- Projects in Machine Learning Research, Spring 2024.

University of Pennsylvania, Philadelphia, Pennsylvania USA
Teaching Assistant

- ENM 360 Introduction to Data-Driven Modelling, Fall 2019-2020.
Responsible for creating homework assignments, creating exams, teaching, and grading.
- MEAM 536 Viscous Fluid Flows Spring 2021.
Responsible for recitations, and grading.

PROJECT SUPERVISION

- **Diego Renner**: "On differentiable simulations of haemodynamic systems", 2023, Co-supervised with Sid Mishra and Ben Moseley, ETHZ, Grade: 6.0, Published.
- **Levi Lingsch**: "Toward Personalized Cardiovascular Healthcare: Flow Matching Neural Operators for Forward and Inverse Problems", 2024, Co-supervised with Sid Mishra, ETHZ, Grade: 6.0, Published.
- **Orestis Oikonomou**: "Formal Grammar Foundation Models for PDE Discovery and Solution", Co-supervised with Sid Mishra and Davide Scaramuzza, UZH, 2024, Grade: 6.0, Published.
- **Karin Yu**: "Formal Language for ODE Discovery", Co-supervised with Eleni Chatzi, ETHZ, Published.

ACADEMIC SERVICE Reviewer for: NeurIPS, ICML, ICLR, Journal of Machine Learning Research, Computer Methods in Applied Mechanics and Engineering, Journal of Scientific Computing, Nature Communications

COMPUTER SKILLS

- ML Frameworks: Jax, Pytorch, NumPy, SciPy, NumPyro, Pandas, Matplotlib, Seaborn.
- Applications: Vi/Vim, Paraview, SimVascular, Mathematica.